My motivation to pursue a PhD in economics grew from a realization during my research in mechanism design and auction theory: proving a new theoretical result is not the same as identifying a question that advances economic understanding. My intensive training in economics and mathematics equipped me to construct models and develop rigorous proofs, but research taught me that meaningful theory also requires judgment: identifying worthwhile questions from economic phenomena and existing theory, using economic intuition to choose appropriate models and methods, and assessing what new insights the results offer. I seek doctoral training from GSB to broaden my theoretical perspective, engage more deeply with frontier research, and sharpen this judgment. My long-term goal is to become an academic economist conducting original research in economic theory.

My studies at Shandong University sparked my interest in microeconomic theory and motivated me to pursue further theoretical training at the University of Queensland (UQ). In mathematics, real analysis, topology, and functional analysis gave me a strong foundation in mathematical analysis and proof. To learn optimization used in economic research, I also studied optimal control and the calculus of variations, static optimization centered on KKT conditions and duality theory, and numerical optimization. Courses in probability theory and stochastic processes further prepared me to analyze uncertainty in economic models. In economics, UQ's Honours program provided PhD-level training in microeconomics, macroeconomics, and econometrics, with courses taken alongside PhD students. With a GPA of 7/7 across all my UQ courses, I was named to the Dean's Honour Roll and was the sole recipient of the Archibald Scholarship. Beyond coursework, I worked as a teaching assistant in mathematics and economics, designed tutorial handouts, and made my lecture notes publicly available. These experiences provided the mathematical tools and theoretical preparation for my research in auction theory and mechanism design.

My first research project in microeconomic theory, supervised by Dr. Allan Hernandez Chanto, studied decentralized team bidding for a scarce, divisible resource. I studied recent working papers on team bidding (Rachmilevitch, 2025; Barbieri et al., 2025) and realized that team bidding arises in many real-world settings and raises many questions worth exploring. After studying auction theory systematically, I decided to begin with a second-price auction. I built a model with predetermined bidding, allocation, and payment rules and focused on the team members' equilibrium bids. While developing the proofs, I identified another novel feature: because members' allocations were private information, their valuations became interdependent, making the equilibrium harder to derive. I therefore read Claudio Mezzetti's *Mechanism Design with Interdependent Valuations: Efficiency* (Mezzetti, 2004) and invited him to join the project. Using fixed-point theory, I derived a closed-form equilibrium under a uniform distribution and obtained several comparative statics results. This experience taught me to draw on the literature to overcome technical difficulties and interpret mathematical results in terms of their economic implications.

My second research project was my honours thesis, jointly supervised by Allan and Claudio. Inspired by Mezzetti (2004), I began to analyse group bidding using a mechanism-design method, shifting focus from equilibrium under predetermined rules to the optimal design of those rules. I built a new model in which a social planner chooses bidding, allocation, and payment rules for a group to maximize welfare subject to budget and incentive constraints. This model has rich practical applications in auctions for land, spectrum, and other rival resources. Unlike in my first project, interdependent valuations did not arise in this setting. I instead used a weighted-surplus approach to derive the optimal mechanism and found that expected cost recovery could require sacrificing efficiency. This raised a question: is this efficiency loss unavoidable, or does it disappear as the group grows? Drawing on the VCG identity in Delacrétaz et al. (2025) and insights on large groups from Birulin (2006), I showed that, for sufficiently large but finite groups, efficiency can be achieved while satisfying the model's constraints. The proof was particularly challenging and required convergence arguments in normed spaces from functional analysis. This experience taught

me to choose models and methods suited to the economic question, draw on the literature for both proof techniques and new research directions, and recognize that identifying worthwhile questions can be as important as proving results.

In another project, I extended my research on team bidding to examine how external auction rules and bidding instruments interact with internal incentives. During the UQ Winter Research Program, under Allan's supervision, I examined real-world transactions and noticed that bidders can offer securities tied to a project's future returns rather than only cash. This led me to ask how internal incentives shape equilibrium bidding when teams of privately informed members compete using securities. Building on the security-bid auction framework of DeMarzo et al. (2005), I developed a model of teams composed of privately informed members competing in a first-price auction using equity securities. I characterized the separating equilibrium in closed form and proved that an internal mechanism implementing the integrated-team equilibrium satisfies incentive compatibility and individual rationality. This experience taught me to identify research questions from real-world economic institutions, translate them into tractable theoretical models, and draw on existing theory and rigorous mathematical analysis to investigate them.

Stanford GSB's PhD program would provide an ideal academic environment for the next stage of my theoretical research. Its rigorous graduate training would deepen my theoretical foundation and broaden my methods for analyzing economic problems. I also hope to develop my research judgment and ability to conduct research through exchanges with faculty and other PhD students. I am particularly interested in **Professor Andrzej Skrzypacz**'s research on auction theory, especially how bidding instruments and auction formats shape equilibrium outcomes. Reading DeMarzo et al. (2005) deepened my understanding of how the properties of securities affect equilibrium bidding and seller revenue. This closely relates to my research on my research on group security auctions and motivates me to explore how security design and internal incentive constraints jointly affect auction outcomes. **Professor Michael Ostrovsky** shows how changing the payment basis from per appearance to per install can alter the composition of auction winners and bidders' subsequent incentives (Ostrovsky, 2023). This has encouraged me to investigate how payment-sharing rules and other institutional details shape bidding behavior, participation, and efficiency in decentralized teams.

I also believe I can contribute to GSB. Drawing on my research experience in microeconomic theory, I look forward to participating in the academic community at Stanford and sharing my ideas with faculty and fellow students. I also hope to contribute to teaching and academic exchange through my experience as a teaching assistant in mathematics and economics. In the long run, I hope to become an academic economist who combines research and teaching, and advances theoretical economics by asking meaningful economic questions and conducting original research.

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